Hackathon - Report

In light of the current situation in our country, we've chosen to include in this document the memes we prepared during the hackathon—before it came to its cruel end.

יחיעם וגילי מחפשים שותף נוסף להקאתון

Part 0: Pre-process the data

We began by normalizing "null" tokens (e.g. "NA", "—") into genuine missing values, then systematically parsed and recast each raw column—splitting composite identifiers, extracting numeric grades or percentages from free-text fields, and mapping Hebrew terms into English categories. Date-time fields were converted into proper date and time types, and clinical markers like HER2 and Ki-67 were processed into both numeric scores and categorical flags. We then one-hot encoded relevant categorical variables (e.g. surgery types, TNM tumor categories), coerced count fields into integers, and dropped the original raw columns. Finally, we wrapped all of this in a pipeline that imputes any remaining missing values and scales continuous features to prepare a clean, fully numeric dataset ready for modelling.

Part 1: Predicting Metastases

Goal

The goal of Part 1 is to predict the presence of metastases across 11 anatomical sites using patient clinical features. Each patient can have metastases in multiple sites simultaneously, framing this task as a multi-label classification problem.

1. Baseline: Per-Label Decision Tree Model

Model Description

We implemented a baseline model by training a separate decision tree classifier for each metastasis site using MultiOutputClassifier. Each label was treated independently. The classifier outputs a score vector, where each entry is the predicted probability of metastasis for that label.

Training Procedure

- Input features were taken from the cleaned.csv dataset.
- The training data was sampled (15%) and split into training\validation sets.
- Missing values were imputed using an ImputerPipeline.
- For each label, a decision tree of fixed depth was trained.
- Binary predictions were obtained using a 0.5 probability threshold.

Evaluation

Predictions were evaluated on the validation set using the provided scoring script, which computes both **micro-F1** and **macro-F1** scores.

Baseline Results

k	micro_f1	macro_f1
3	0.33	0.09
5	0.424	0.205
10	0.64	0.38
15	0.72	0.42
20	0.737	0.43

2. Advanced Model: Neural Network with Decision Tree Scores

Model Description

To enhance prediction performance, we introduced a neural network model that leverages the decision tree outputs. Specifically:

- Train a decision tree for each label to generate a score vector (sama as in the baseline). Unlike the baseline, the output is not binarized; instead, we retain the raw predicted probabilities for each metastasis site.
- This score vector is concatenated with the original input features.
- The combined vector is then fed into a feedforward neural network (BoostedMetastasisNN) for final multi-label classification.

This design allows the network to learn higher-order interactions between features and site-specific cancer risk signals.

Training Procedure

- Missing values were imputed using the same ImputerPipeline as before.
- Decision trees were trained as in the baseline and ”frozen” to produce score vectors.
- The score vectors were concatenated with the original features.
- A simple neural network was trained on the combined inputs.
- Model optimization was done via Adam with binary cross-entropy loss.

Hyperparameter Selection

We performed grid search cross-validation over the following hyperparameters:

- Depth (the tree depth)
- n_epochs
- batch_size
- learning_rate

For each combination, we trained the full pipeline and selected the configuration with the highest average F1 score across validation folds.

Evaluation

The best-performing model (based on CV) was evaluated on the held-out validation set using the same scoring script.

Advanced Model Results

Micro f1 = 0.9783420463032113

Macro f1 = 0.8603153715522881

Part 2: Predicting Tumor Size

1. **Data Processing:** We engineered and cleaned features by transforming numeric variables to capture non-linear effects, one-hot encoding categorical fields, imputing missing values, and selecting the most informative predictors—ensuring the model sees only the strongest, noise-reduced signals.

2. **Data Split:** The dataset was partitioned into training, validation, and test sets so that model fitting, hyperparameter tuning, and final evaluation each occur on disjoint samples, guaranteeing an unbiased estimate of performance.
3. **Baseline XGBoost:** A gradient-boosted tree model was trained first to capture complex, non-linear interactions among clinical and demographic variables, laying a robust foundation for downstream learning.
4. **Stacking Predictions:** We then augmented the training data with the XGBoost model's predictions as an extra feature, allowing our neural network to leverage those insights in its own pattern recognition.
5. **Neural-Net Tuning:** Through grid search and early stopping, we optimized the network's architecture, learning rate, and regularization strength to maximize predictive accuracy while preventing overfitting.
6. **Final Model Training:** With the best hyperparameters locked in, we retrained on the combined training+validation data and assessed performance on the held-out test set to obtain our definitive evaluation metric.
7. **Test-Set Predictions:** Finally, we applied the entire preprocessing and modeling pipeline to the unseen test file, producing and saving high-quality predictions ready for downstream analysis or submission.

Results – Baseline

MSE = 1.328823

Results - Model

MSE = 0.35235190201955124

Part 3: Unsupervised Data Analysis

We split the team for 2 sub-teams and each worked on different direction:

First:

We perform an unsupervised analysis on features to identify biologically meaningful subtypes based on hormone receptor expression and tumor morphology, and characterize each subtype across additional clinical variables.

Methods

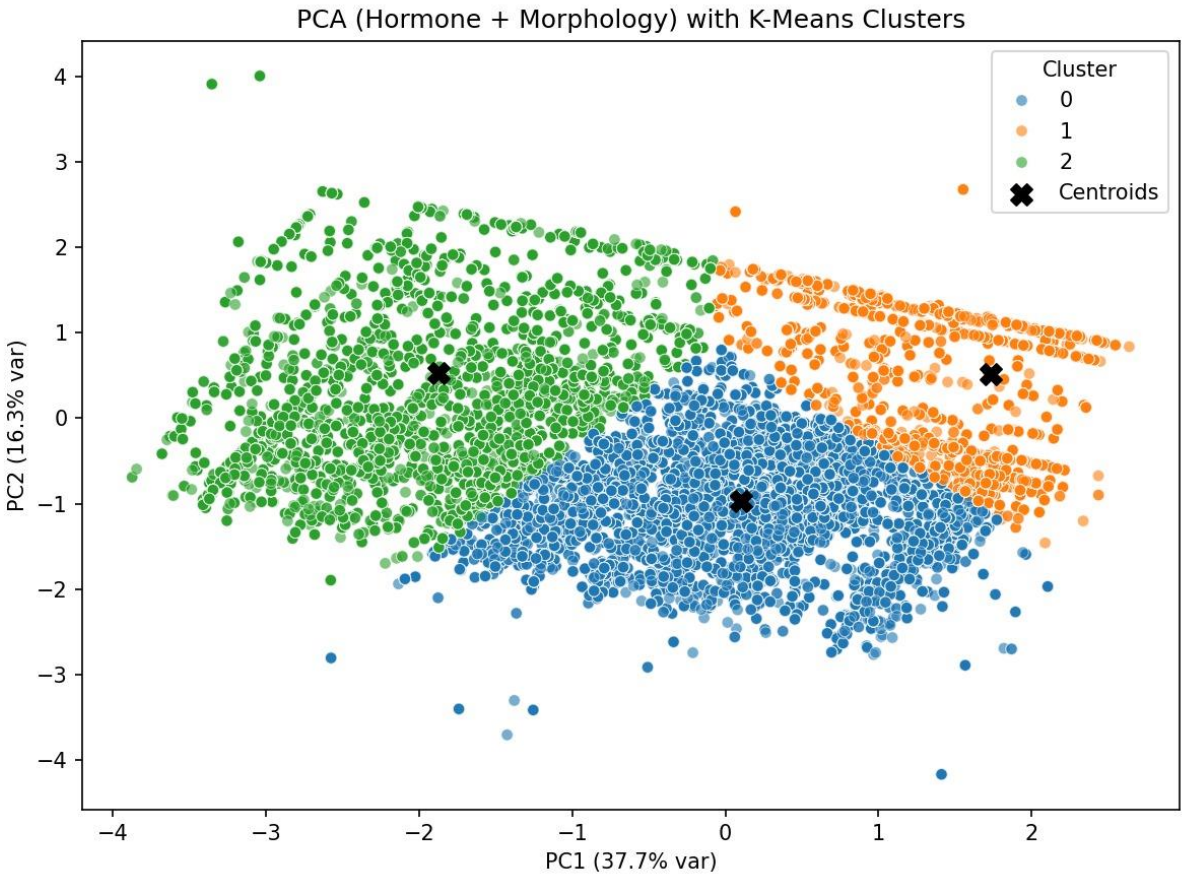
We began by importing the data into a pandas DataFrame and standardizing the column names to ensure consistency. We removed all administrative and date-related columns (e.g., IDs, hospital codes, surgery timestamps) that do not directly reflect tumor biology or patient characteristics.

Next, we identified seven key biological features that capture hormone status and tumor morphology: ER, PR, HER2, KI67, and three measures of tumor size and stage (width, depth, TNM T-stage).

To reduce dimensionality and emphasize the dominant axes of variation, we applied PCA to the seven standardized features. We retained the first two principal components, which together explained approximately 54.0% of the total variance (37.7% for PC1 and 16.3% for PC2). This projection allows us to visualize the most informative directions in the data space.

Finally, we used K-Means clustering on the two-dimensional PCA embedding to partition patients into three subtypes (k=3). This unsupervised step groups patients according to similarity in their combined hormone and morphology profiles, revealing a spectrum from high-receptor/low-size tumors to low-receptor/high-size tumors, with an intermediate group.

Results

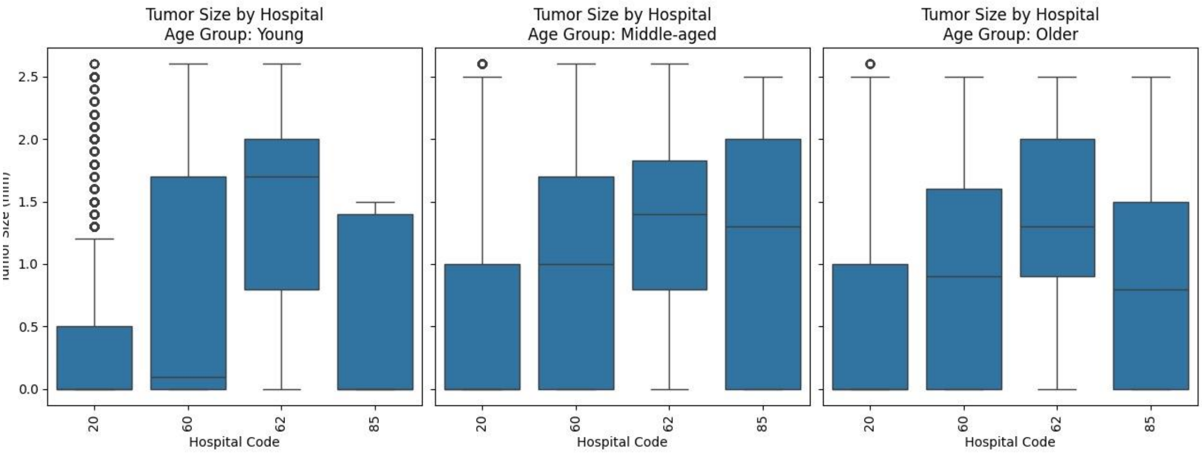


The three clusters separate along this continuum:

- **Cluster 1** (orange points, rightmost) has the **highest hormone marker levels** and **smaller tumors**, indicating a hormone-driven subtype with lower tumor burden.
- **Cluster 2** (green points, leftmost) shows the **lowest receptor levels** and **larger, more advanced tumors**, consistent with a non-hormone, aggressive phenotype.
- **Cluster 0** (blue points, center/bottom) falls **intermediately** on both axes, blending moderate hormone expression with intermediate tumor size.

We didn't find some unique conclusion from the analysis we made, but we do think that perhaps further analysis with features we don't have here can find some interesting characteristics of each group.

Second:



ניתן להסיק מהתרשים:

- קבוצת גיל – Young:

בית חולים 20: גידולים קטנים במיוחד. החציון קרוב ל־0.5 מ״מ, רוב התצפיות נמוכות.

בתי חולים 60 ו־62: גידולים גדולים יותר באופן משמעותי. החציון סביב 1.5–2 מ״מ.

בית חולים 85: באמצע, מעט נמוך יותר מ־62.

יש שונות ניכרת בין בתי החולים, גם לאחר סילוק האאוטליירים.

- קבוצת גיל – Middle-aged:

כל בתי החולים מציגים חציון בין 1 ל־1.7 מ״מ.

בית חולים 62 ממשיך להיות גבוה יחסית (בדומה לקבוצת הצעירים).

הפערים מצטמצמים בהשוואה לקבוצה הצעירה.

הבדלים קיימים אך מתונים יותר. ייתכן שהגיל מפחית את ההבדלים בין המוסדות.

- קבוצת גיל – Older:

החציון בבית חולים 62 ו־60 נשאר מעט גבוה יותר.

בית חולים 85 מציג גידולים קטנים יותר באופן עקבי.

הפיזור בכל קבוצות הגיל רחב, אך ללא קפיצות קיצוניות.

נראה שבגיל המבוגר — ההבדלים בין בתי החולים קטנים יותר.

לסיכום, לאחר סילוק האאוטליירים, ניתן לראות תמונה נקייה ואמינה יותר של ההבדלים האמיתיים בין בתי החולים.

בית חולים 62 נוטה להציג גידולים גדולים יותר בכל קבוצות הגיל.

בית חולים 20 מתבלט בגידולים קטנים מאוד אצל צעירות, אך מתקרב לאחרים עם העלייה בגיל.

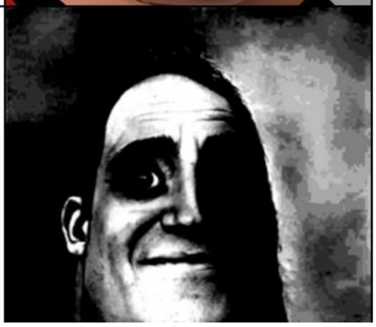
השונות מצטמצמת ככל שהגיל עולה, ייתכן שקשור למדיניות אחידה יותר בגיל מבוגר או לגילוי מוקדם יותר בצעירים במקומות מסוימים.

Summary

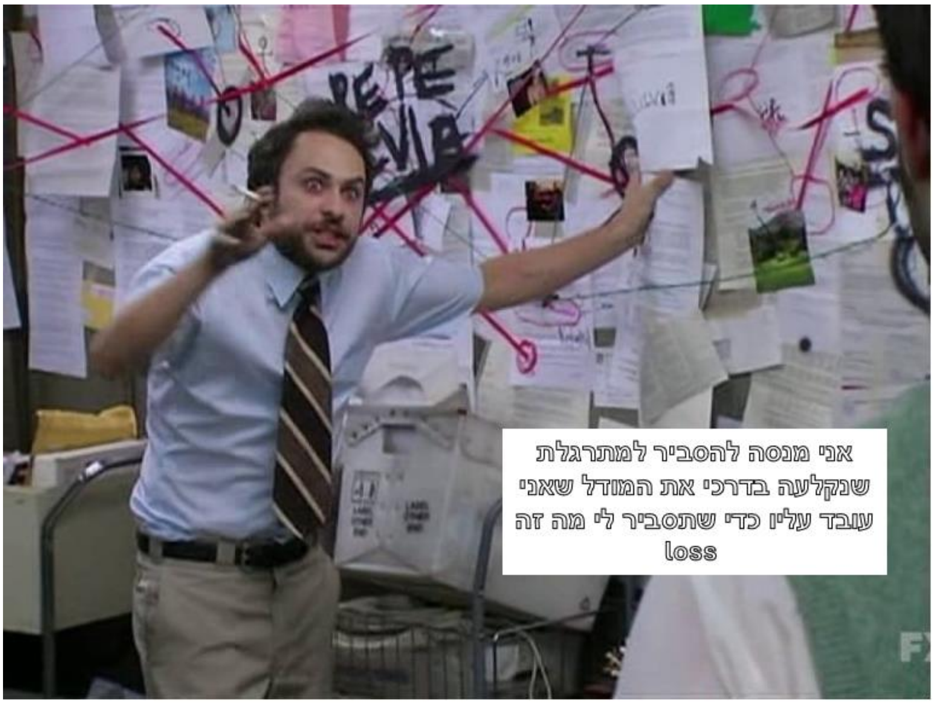
We had a great time! Here are the rest of the memes:

אני כשהגיעה שעת חצות והכנתי כבר שישה מימז אבל המודל עוד לא רץ

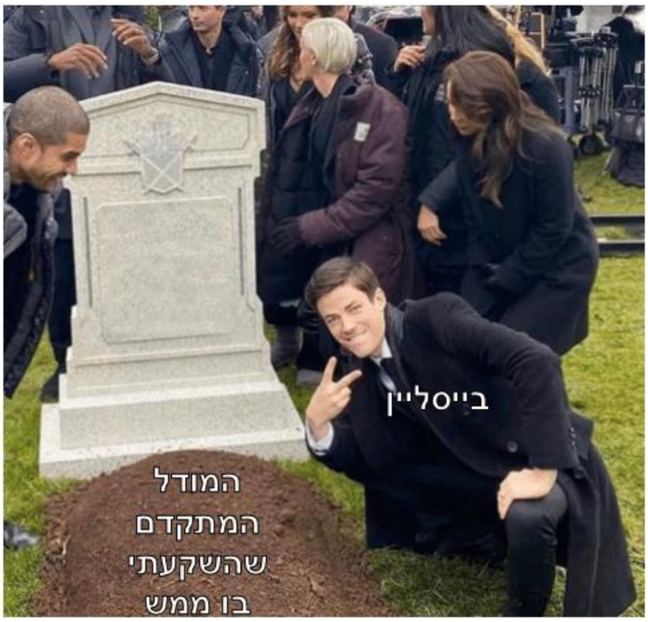
אני בהפסקת סיגריה לפני שאני חוזרת לקבוצה ומספרת להם שהמודל המפותח שעבדתי עליו קרס

12.06.25 **9:00**	
13.06.25 **9:00**	

אני, כשהקבוצה שלי חוזרת מהפסקה:



Our personal favorites:



תגובתי לחבר הצוות שלי כשהוא אומר שהוא חושש שייקח למודל עוד הרבה זמן לסיים והשעה כבר 2 לפנות בוקר